Shenzhen Dudumiao Technology Co., Ltd.

smokefire Fire, Smoke & Smoking Detection

Model Capability Test Guide

Document version: V1.0
Applicable software version: 1.0.0
Updated: 2026-08-03

Contents

Important
This guide uses real images from public datasets to show actual model output and practical boundaries. smokefire is a video-AI assistance tool, not a certified fire alarm and not a substitute for statutory fire protection, patrols, emergency procedures, or human verification.

Document information

Item	Value
Models	fire_smoke_v5.pt, smoking_v4.pt
Default threshold	0.30 for every image in this guide
Input size	640
Test sample	18 D-Fire images plus 12 CigDet images; 30 total
Runtime	NVIDIA CUDA validation environment with PyTorch 2.9.1+cu128 and Ultralytics 8.4.24; the validation-host model does not define customer GPU compatibility
Project	https://github.com/newtv-ai/smokefire

1. Summary

Fixed sample group	Images	Actual result at 0.30	Interpretation
D-Fire fire-only	5	Fire boxes on 5 images	Responded to visible flames and several distant/night fire points
D-Fire smoke-only	5	Smoke boxes on 2; no detection on 3	Distant, thin, or low-contrast smoke remains a clear boundary
D-Fire fire-and-smoke	4	At least one class on all 4; both classes on 3	Fire and smoke are evaluated separately in one scene
D-Fire neither	4	No box on 3; smoke 0.49 on 1	Clouds, haze, glare, and similar patterns can cause false alerts
CigDet positive smoking	12	Smoking boxes on all 12; top scores from 0.35 to 0.94	Stronger on close views with visible hand/cigarette/mouth relationships

These counts describe only the 30 fixed images shown here. They are not overall accuracy, recall, false-alert rate, or a project guarantee. Public datasets can differ materially from the target cameras.

2. Data sources and method

2.1 Public sources

Dataset	Use	Official location	License
D-Fire	Fire, smoke, combined, and neither images	https://github.com/gaia-solutions-on-demand/DFireDataset	CC0 1.0
CigDet	Real smoking/cigarette scenes	https://data.mendeley.com/datasets/6hyrr8typ7/1, DOI 10.17632/6hyrr8typ7.1	CC BY 4.0

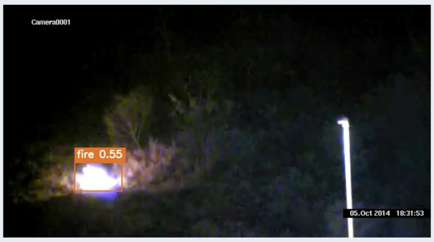
D-Fire images were read from samples tagged dfire in the public mirror at https://huggingface.co/datasets/Hajorda/flameye-wildfire-detection; the official source and license remain the D-Fire repository above. CigDet is credited to Ali Khan, and this guide uses images from its public test set.

2.2 Fixed selection and inference

- D-Fire images were sorted by public image ID, then evenly sampled as 5 / 5 / 4 / 4 from fire-only, smoke-only, combined, and neither groups.
- CigDet test images were sorted by filename, then 12 images were selected at even intervals from the 111-image test set.
- Both delivered models used input size 640 and the default 0.30 threshold. Only model-returned boxes at or above 0.30 were drawn.
- No box was selected or added by hand. Misses remain unboxed and false positives remain visible.

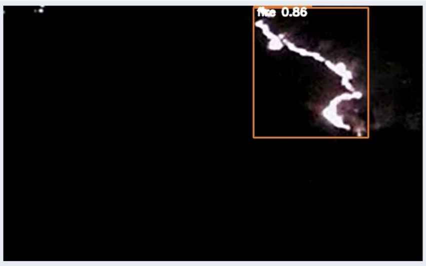
> **Important**
> Images were downloaded from real public datasets and may differ from the target site. Every box was generated by actual inference with fire_smoke_v5.pt or smoking_v4.pt; no box was added manually, and confidence is displayed to two decimal places.

3. Fire results



D-Fire F01

D-Fire F02



D-Fire F03

D-Fire F04

D-Fire F05

D-Fire fixed fire-only samples F01-F05

ID	Public image ID	Highest actual output
F01	dfire_AoF00001	fire 0.55
F02	dfire_AoF04540	fire 0.71
F03	dfire_AoF04591	fire 0.86
F04	dfire_PublicDataset00007	fire 0.64
F05	dfire_PublicDataset00358	fire 0.42

All five produced a box. F05 reached only 0.42, showing that confidence and stability can fall for distant small targets.

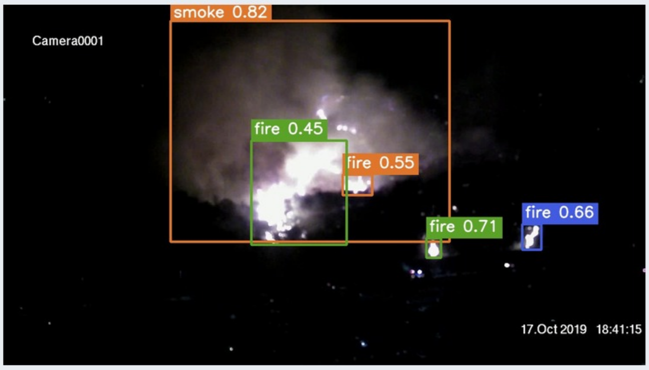
4. Smoke results

D-Fire fixed smoke-only samples S01-S05

ID	Public image ID	Highest actual output
S01	dfire_AoF00002	No detection
S02	dfire_AoF06061	No detection
S03	dfire_PublicDataset00162	No detection
S04	dfire_PublicDataset00469	smoke 0.81
S05	dfire_PublicDataset00648	smoke 0.93

Large plumes with clear boundaries and background contrast scored strongly. Distant, thin, low-contrast smoke or smoke that resembles cloud/haze can be missed.

5. Combined fire/smoke results



D-Fire B01

D-Fire B02

D-Fire B03

D-Fire B04

D-Fire fixed combined samples B01-B04

ID	Public image ID	Main actual outputs
B01	dfire_AoF04019	smoke 0.82; fire up to 0.71
B02	dfire_AoF04467	fire 0.63; smoke not detected
B03	dfire_AoF07841	fire 0.76; smoke 0.56
B04	dfire_PublicDataset00656	smoke 0.97; fire 0.55

All four produced at least one class, and three produced both. B02 shows that a combined scene does not guarantee that both classes pass the threshold.

6. No-fire/no-smoke interference

D-Fire N01

D-Fire N02

D-Fire N03

D-Fire N04

D-Fire fixed neither samples N01-N04

ID	Public image ID	Highest actual output
N01	dfire_AoF00009	No box
N02	dfire_AoF01530	smoke 0.49
N03	dfire_AoF03575	No box
N04	dfire_AoF07717	No box

N02 is a retained real false-positive example. Site tests should include clouds, fog, steam, dust, reflections, welding, and night lighting rather than positive images alone.

7. Smoking results

7.1 C01-C04

CigDet C01

CigDet C02

CigDet C03

CigDet C04

CigDet fixed samples C01-C04

ID	Source file	Highest actual output
C01	smoking_0020.jpg	smoking 0.72
C02	smoking_0068.jpg	smoking 0.55
C03	smoking_0122.jpg	smoking 0.85
C04	smoking_0196.jpg	smoking 0.94

7.2 C05-C08

CigDet C05

CigDet C06

CigDet C07

CigDet C08

CigDet fixed samples C05-C08

ID	Source file	Highest actual output
C05	smoking_0254.jpg	smoking 0.89
C06	smoking_0330.jpg	smoking 0.75
C07	smoking_0374.jpg	smoking 0.37
C08	smoking_0413.jpg	smoking 0.81

7.3 C09-C12

CigDet C09

CigDet C10

CigDet C11

CigDet C12

CigDet fixed samples C09-C12

ID	Source file	Highest actual output
C09	smoking_0451.jpg	smoking 0.35
C10	smoking_0484.jpg	smoking 0.78
C11	smoking_0528.jpg	smoking 0.57
C12	smoking_0559.jpg	smoking 0.40

Every positive image produced a box at or above 0.30, but C07, C09, and C12 topped out at only 0.37 / 0.35 / 0.40. Smoking detection is sensitive to cigarette size, hand occlusion, pose, and distance. These close-view results must not be extrapolated directly to distant surveillance views.

8. Capability boundaries

- The model can analyze only what the camera captures. Blind spots, occlusion, overexposure, darkness, and compression loss cannot be fully recovered.
- Distant flames, cigarettes, and thin smoke may contain too few pixels even in a nominally high-resolution stream.

- Smoke is especially sensitive to clouds, fog, steam, dust, and background brightness; this fixed sample contains both misses and a false positive.
- CigDet is dominated by close views, while production CCTV is often high-angle and distant. Distribution shift directly changes results.
- Confidence is a relative model score for a candidate, not the probability that the real-world event is true, and scores are not comparable across models.
- A still image validates only one frame. Video alerts also depend on sampling interval, consecutive-frame policy, stream continuity, and compute load.

9. Site acceptance

1. Build positive, ordinary-negative, and hard-negative samples for every representative camera view.
2. Cover day, night, backlight, rain/fog, distance, bitrate, and camera-angle variation.
3. Establish a baseline at the default `0.30` threshold. If it changes, replay the same set and measure both misses and false alerts.
4. Score whether a video event forms and record time-to-first-alert and evidence completeness.
5. Preserve software version 1.0.0, both model hashes, sample list, and results for reproducibility.
6. Keep human verification during pilot operation and expand only after agreed acceptance criteria are met.

Prepared by: Shenzhen Dudumiao Technology Co., Ltd. Project: https://github.com/newtv-ai/smokefire